

Ines Hipolito: “Dysfunctional Markov Blankets: From Stuck States to Adaptive Perturbations”

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all right the next video is 23 minutes long in s hippolito dysfunctional markof blankets from stuck states to [Music] adaptation hi everyone I'll be presenting today uh dysfunctional Mark of blankets from Stu states to adaptation this a paper that has been um in progress for quite some time and it is um a collaboration with maasa Alex Kea and Daniel fredman so I want to start by uh highlighting that there are stuck systems or stuck States all around us and that we find that example in neurocognitive um levels such as in OCD patterns in PTSD trauma Cycles or in addiction circuits then on the social or ecological scale uh we find it for example in poverty traps in each chambas and Market uh monopolies and climate tipping points then in AI we also find them for example in a large language models Loops who isn't been stuck in a I apologize kind of cycle then uh in reinforcement learning local Minima in recommendation filter Bubbles and in neural networks training fails so what um these kinds of systems uh share is that uh they are some sort of critical threshold uh they have um or present um self reinforcing loops and uh they have this persistance to uh change so they tend to remain in the same state hence why they are stuck so then the core problem is going to be a sort of a paradox so um the view is that systems adaptive or complex systems um coupled with the environment they should adapt but sometimes they don't so then are from this perspective a few questions um become Salient why do flexible systems become so rigid and do not have that flexibility to adapt to the environment how do these adaptive boundaries uh between the system and the environment break down and what maintains that dysfunctional State what is there that must be so strong that um the system cannot uh change to a different uh or look for or act for uh to attain a different more preferable kind of state so then of course that these presents um these are presented as Challengers because there is an everchanging world and the system that needs to adapt to that everchanging World um and uh it seems like um existing approaches um such as dynamical systems theory and network uh science um still have some limitations to quantify and address this kind of phenomenon that we call a dysfunction so then from our

perspective uh we need um a frame work that unifies that uh unifies this explanation so we need three things first we need to identify and characterize D functional States then we need to provide a mechanistic explanation for the Stu States and finally we need to offer a principled approach to designing these kinds of interventions and that's what we set up to do uh in this paper and that I'll be presenting to you so um our framework Bridges um theoretical Frameworks uh from complex systems theory and the free energy principle the main or core um Concepts uh they are going to be useful are Notions such as emergence self-organization and adaptation from complex systems theory and coupling self-maintenance and free energy from the free energy principle by uh utilizing these two Frameworks uh we developed uh what a construct that we uh defined as dysfunctional Mark of blankets and with that uh we hope that we will offer a formal theory of uh systems dysfunction and we hope to develop quantifiable measures for this phenomenon of stuckness and finally practical recovery strategy so that we can act upon uh the system that is stuck in a therapeutic way and we uh uh hope to kick the system out of that stuck State into a healthy State okay so is adaptive systems we all know that um they are comprised by components which interact and from that interaction we observe some sort of emergence um for example in the case of an immune response we have different immune cells each one with a specific role and they interact in the form of cell signaling Rec recognition of events and that interaction then gives rise to emergence that we observe in the form of coordinated defense or system level memory then uh from the free energy principle which we uh are very acquainted with uh in this conference uh we learn that systems that exist act to minimize their free energy and um further to that um the understanding or the Insight is that systems are couple with the environment and they act upon the environment to precisely uh attain or look for more preferable States for their self-maintenance and we can explain um that behavior through um active inference then another very useful uh construct is Mark of blankets of course the framework is a dysfunctional Mark of blankets and the mark of blanket uh as we all know is a statistical boundary between what is inside of the system and what is outside but further to that uh an important uh Insight is the conditional Independence between internal and external states which give um the uh system sort of a very important boundary um to have a distinction between the self and the no self and that can be seen also through the notion of it's an operationally closed system A system that self organizes and maintains itself for that maintenance it's also quite important that it is thermodynamically open it is an open system it's not a closed system and that's what allows them to remain alive by adjusting to the environment and here you see for example we have a neurons in the form of internal states

that could be part of a network so a key Innovation concept that we bring in is the notion of selectability and selectability is a systems capacity to explore navigate and select among possible States these can also be seen as the system having multistability many points um that are afforded to the system in the environment such that uh the system can be healthy in that navigation so you can see that for example that comes represented here in the form of state A B C or D different points that the system can explore within the environment and select the most preferable ones and um the notion of selectability allows to um think about State space accessibility or affordances the adaptation capacity and the boundary flexibility to adapt rather than being rigid and stuck in a particular state so then uh within the free energy principle we can think through that Paradox it becomes if systems minimize free energy if they Accord to the principle why would they get stuck does the principle not work or is it the system that is uh not in a stuck state so then uh what we realize is that um what is not what is really within the Paradox is that we cannot explain this persistent maladaptive States and there is not really a formal treatment of the dysfunction so then there is a missing link between Theory and intervention and our solution is the dysfunctional Mark of blankets there are two types of boundary dysfunction that we can observe uh in this systems that are not adjusted attuned adapt to the environment so they can come in the form of stock States or blanket breakdown and stock States I mentioned this in the first slide and uh for example in the case of depression what we observe is rigid kinds of thought patterns reduced emotional flexibility impaired Social uh information processing and shortcuts bypassing healthy emotional regulation and in a more General way we can Define these stck states as the self reinforcing kind of Cycles being trapped in deep energy Wells and being resistant to environmental perturbation then on the other hand we have blanket break down and an example of this is the immune system when the immune system attacks its own tissue and what happens here is a very interesting phenomenon of excessive permeability um on the boundary of the blanket right loss of system Integrity so the self no self-d distinction the system doesn't know what's self and what is not self that's the loss of of system Integrity because of the boundary dissolution okay so then we arrive at uh the definition of the dysfunctional Mark of blankets and this this is a statistical boundary that um contrastingly has lost its adaptive capacity for Effective mediation between internal and external States and the key characteristics of um the dysfunctional Mark of blankets are that the system is persistent in High free energy despite adaptation attempts and what we observe is the breakdown in information processing and the loss of effective environmental coupling I will unpack this further but let me just situate us um for a moment so

uh we will go from the framework to practical tools to cross scale applications that's where we would like to go uh so to do that we need to First Define quantifiable measures then um identify a dysfunction index and finally the recovery metrics then as to practical tools we need adaptive perturbations in order to kick um the system out of the valley and Recovery strategy and this on a more therapeutic kind of site and hopefully we'll be able to apply these cross across scales so now looking at the framework itself the first thing that we need to do is to um find that dysfunction index and what that is going to help us do is to measure and identify and delineate the dysfunction it will measure the information flowing through the wrong channels like a diagnostic kind of tool that um spots the system bypass and the higher Dev value the more dysfunction for example like a cell with a compromised membrane then uh we will need also a recovery metric and this will be a pro a progress kind of indication indicator for us so it shows a dysfunction reduction and the thought here is is that um we ask ourselves are we fixing the bypass problem so then uh we will need uh to uh in order to advance with that we need the landscape uh reshaping kind of like bringing out a new geography of points uh that are afforded to the system making sure that we don't get stuck again and hopefully we'll attain the systems adaptability where we find the question can it be a explore can it explore and respond flexibly to uh the environment so why are these two measures relevant well they are they will uh serve the role of being Universal tools across scales we will be able to apply them uh to uh all scales from cells to ecosystems and they will help us diagnose and track recovery and guide our next stage which is the intervention design okay so now moving to practical tools adaptive perturbations and Recovery strategies uh we have adaptive perturbations here playing a crucial role and um an Adaptive perturbation is going to be a controlled intervention that temporarily disrupts a system state to facilitate adaptive change and what we mean by this or our goal is uh not simply to push a system out of being stuck but it is to help it to become more resilient precisely to make sure that it doesn't come back to the stuck State as you can see an illustration here on the graph below so uh then we Define the Adaptive perturbation design as in the first term as traditional intervention approach where we would like to nudge the system towards uh these better States such that it adapts based on the system's response then with the middle term we can reshape the energy landscape have a different sort of more um flexible uh geography that makes it harder to get stuck again like smoothing out the deep valleys and then finally uh we can promote exploration and that means help the system uh discover new affordances new possibilities hence enhance adaptability um there are kind of like two examples of adaptive perturbations that um I just want to mention and uh one is by

the parameter changes where for example you can think of that as a temperature modification in cellular environments and the other one is the external force or input an example of that is trans cranial uh magnetic uh stimulation so why is this adaptive perturbation relevant well it combines three intervention strategies self adjusting all parameter parameters will adapt over time um and it uh includes a balance between immediate Improvement long-term perturbation sorry long-term prevention and future adaptability the Practical impact of um an Adaptive pupation is that it will help us guide for Designing interventions and work across different scales and prevent the kind of uh so to speak Band-Aid Solutions right so then why do ppts matter in the end well for the stuck States they're going to be very relevant to help us overcome energy barriers um break self reinforce in cycles and enable the system to have more States and the capacity to navigate um and explore uh the the environment and select the states that are going to be much more preferable to the continuation and maintenance and self-maintenance and self organization of the system itself in the case of the blanket breakdown which is this is the case of for example the immune system that has lost uh the capacity to differentiate between what is self and what is no self and attacks um its own tissue then utilizing an Adaptive perturbation uh will be uh useful to restore boundary function to stabilize information flow and to reset System Dynamics and therefore uh repair uh the capacity of the system ultimately to recognize what's self and what is no self what is internal States what is external States okay so uh I will recap uh the journey now uh from the dysfunction to the recovery um so first we need to measure the dysfunction then we need to track the recovery then we will need to design interventions and we do that through the Adaptive uh perturbation to implement the recovery finally we need to monitor that kind of recovery such that we now have um been able to uh get the system to be adapted to the environment uh rather than uh just um in a very short term so key relationships that come up within this journey include going from higher dysfunction to Stronger perturbation that is required going to better recovery to reduce intervention strength and going from more selectability to Greater adaptation capacity future directions uh where we need to go is to extend uh this mathematical formalism to non erotic systems to multiple time scale Dynamics and coupled systems interactions in order to do that we also need to uh develop measurement tools such as realtime dysfunction metrics early warning indicators uh that can predict a Tipping Point or a point of no return and then we can apply a preventive measure and cross scale validity tests some of the implemental implementation challenges that um we can see um include of course measurement intervention validation and scaling up and in order to address them we will need uh better tools more

precise tools standard tests and infrastructure the potential impact of uh this framework um is like I said across scales uh in healthcare for example we can look into uh Therapeutics for uh any sort of disease kind of Cycles uh it includes uh mental health and aging patterns in climate action uh we can apply um this uh framework to tipping points to as a preventive measure and to Restoration in AI we can apply to uh develop more robust AI we can also apply it to uh make more smoother ethical and governance uh measures for safety in AI as well as um can help us design bioinspired AI such that it's a system that is uh more adapted uh and hence more effective finally in Social systems we can also apply framework for uh policy design for economical stability and social cohesion and um coming to the end uh call for Action um so uh research priorities include developing standard standardized standardized metrics and build shared data sets create open-source tools and establish cross disciplinary teams so please join us in uh theory development tool creation application testing and knowledge sharing conclusion and takeaway so we went in this journey from uh these functional systems that we observe that are stuck in a local minimum and we want to develop a perturbation in order to help the system become more resilient and adapt to the environment out of the stuck State and enhance a full recovery in the long term so then key takeaways uh are that we hope that we um have developed the foundations for a formal theory of system dysfunction quantifiable measures for the stuckness phenomenon and a practical recovery strategy that can be applied that can be further in the future applied to different scales thank you so much for your time please do not hesitate to contact myself or any of the authors uh if you'd like to uh engage or learn more about uh these work